

CSci 127: Introduction to Computer Science



hunter.cuny.edu/csci

Today's Topics



- Recap: Decisions
- Logical Expressions
- Circuits
- Binary Numbers

Today's Topics



- **Recap: Decisions**
- Logical Expressions
- Circuits
- Binary Numbers

Challenge

Some challenges with types & decisions:

```
#What are the types:
```

```
y1 = 2017
y2 = "2018"
print(type(y1))
print(type("y1"))
print(type(2017))
print(type("2017"))
print(type(y2))
print(type(y1/4.0))
```

```
x = int(y2) - y1
if x < 0:
    print(y2)
else:
    print(y1)
```

```
cents = 432
dollars = cents // 100
change = cents % 100
if dollars > 0:
    print('$'+str(dollars))
if change > 0:
    quarters = change // 25
    pennies = change % 25
    print(quarters, "quarters")
    print("and", pennies, "pennies")
```

Python Tutor

```
#What are the types:
```

```
y1 = 2017
```

```
y2 = "2018"
```

```
print(type(y1))
```

```
print(type("y1"))
```

```
print(type(2017))
```

```
print(type("2017"))
```

```
print(type(y2))
```

```
print(type(y1/4.0))
```

```
x = int(y2) - y1
```

```
if x < 0:
```

```
    print(y2)
```

```
else:
```

```
    print(y1)
```

(Demo with pythonTutor)

Decisions

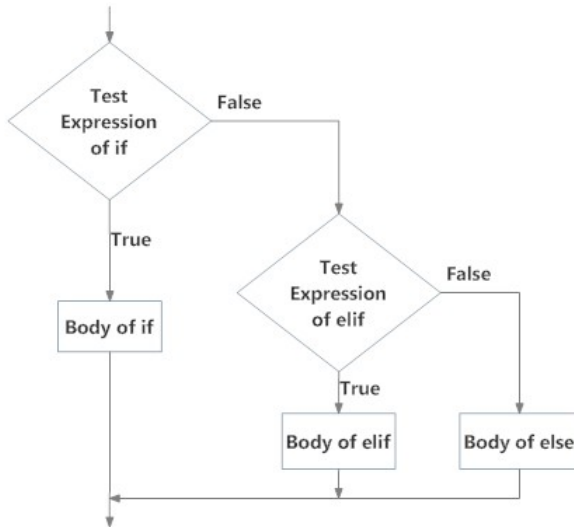
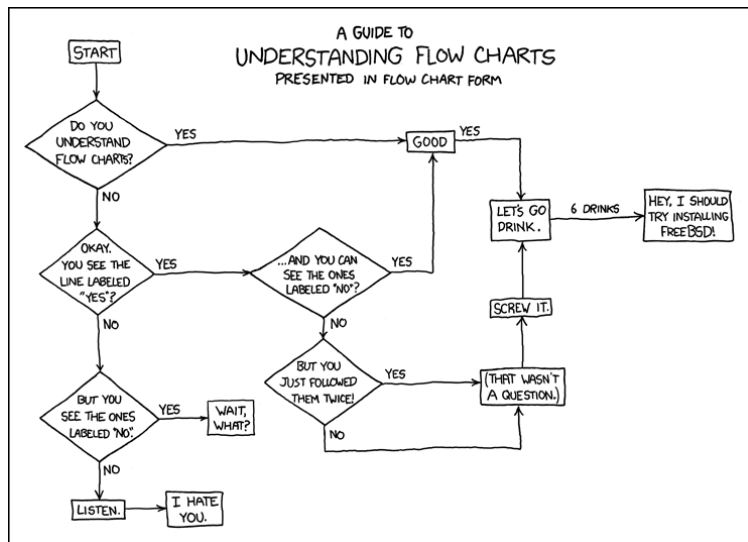


Fig: Operation of if...elif...else statement

Side Note: Reading Flow Charts



(xkcd/518)

Today's Topics



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Challenge

Predict what the code will do:

```
origin = "Indian Ocean"
winds = 100
if (winds > 74):
    print("Major storm, called a ", end="")
    if origin == "Indian Ocean" or origin == "South Pacific":
        print("cyclone.")
    elif origin == "North Pacific":
        print("typhoon.")
    else:
        print("hurricane.")

visibility = 0.2
winds = 40
conditions = "blowing snow"
if (winds > 35) and (visibility < 0.25) and \
    (conditions == "blowing snow" or conditions == "heavy snow"):
    print("Blizzard!")
```

Python Tutor

```
origin = "Indian Ocean"
winds = 180
if (winds > 74):
    print("Major storm, called a ", end="")
    if origin == "Indian Ocean" or origin == "South Pacific":
        print("cyclone.")
    elif origin == "North Pacific":
        print("typhoon.")
    else:
        print("hurricane.")

visibility = 0.2
winds = 40
conditions = "blowing snow"
if (winds > 35) and (visibility < 0.25) and \
    (conditions == "blowing snow" or conditions == "heavy snow"):
    print("Blizzard!")
```

(Demo with pythonTutor)

Logical Operators

and

in1		in2	<i>returns:</i>
False	and	False	False
False	and	True	False
True	and	False	False
True	and	True	True

Logical Operators

and

in1		in2	<i>returns:</i>
False	and	False	False
False	and	True	False
True	and	False	False
True	and	True	True

or

in1		in2	<i>returns:</i>
False	or	False	False
False	or	True	True
True	or	False	True
True	or	True	True

Logical Operators

and

in1		in2	returns:
False	and	False	False
False	and	True	False
True	and	False	False
True	and	True	True

or

in1		in2	returns:
False	or	False	False
False	or	True	True
True	or	False	True
True	or	True	True

not

	in1	returns:
not	False	True
not	True	False

Challenge

Predict what the code will do:

```
semHours = 18
reqHours = 120
if semHours >= 12:
    print('Full Time')
else:
    print('Part Time')

pace = reqHours // semHours
if reqHours % semHours != 0:
    pace = pace + 1
print('At this pace, you will graduate in', pace, 'semesters,')
yrs = pace / 2
print('(or', yrs, 'years).')

for i in range(1,20):
    if (i > 10) and (i % 2 == 1):
        print('oddly large')
    else:
        print(i)
```

Python Tutor

```
semHours = 18
reqHours = 120
if semHours >= 12:
    print('Full Time')
else:
    print('Part Time')

pace = reqHours // semHours
if reqHours % semHours != 0:
    pace = pace + 1
print('At this pace, you will graduate in', pace, 'semesters,')
yrs = pace / 2
print('Or', yrs, 'years'.)
```



```
for i in range(1,20):
    if (i > 10) and (i % 2 == 1):
        print('oddly large')
    else:
        print(i)
```

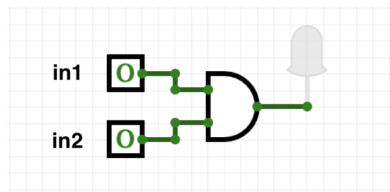
(Demo with pythonTutor)

Today's Topics



- Recap: Decisions
- Logical Expressions
- **Circuits**
- Binary Numbers

Circuit Demo

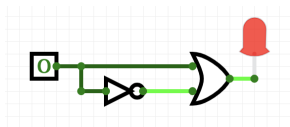


(Demo with `circuitverse`)

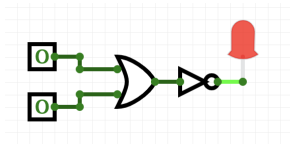
Challenge

Predict when these expressions are true:

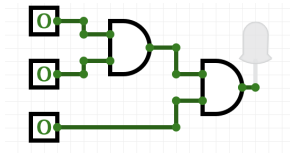
- `in1 or not in1:`



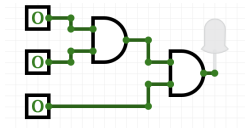
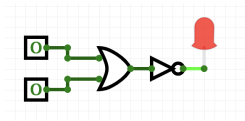
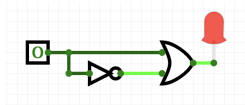
- `not(in1 or in2):`



- `(in1 and in2) and in3:`

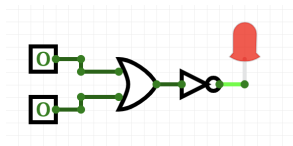


Circuit Demo



(Demo with circuitverse)

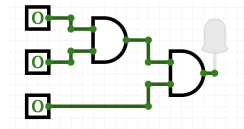
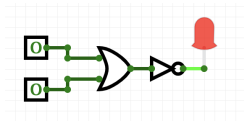
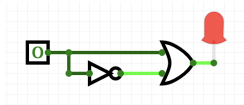
Challenge



Draw a circuit that corresponds to each logical expression:

- $\text{in1} \text{ or } \text{in2}$
- $(\text{in1} \text{ or } \text{in2}) \text{ and } (\text{in1} \text{ or } \text{in3})$
- $(\text{not}(\text{in1} \text{ and } \text{not } \text{in2})) \text{ or } (\text{in1} \text{ and } (\text{in2} \text{ and } \text{in3}))$

Circuit Demo



(Demo with circuitverse)

Today's Topics



- Recap: Decisions
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- **Binary Numbers**

Binary Numbers

- Logic $\rightarrow$ Circuits $\rightarrow$ Numbers

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- Digital logic design allows for two states:

Binary Numbers

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 - ▶ On / Off (two voltage levels)

Binary Numbers

- Logic $\rightarrow$ Circuits $\rightarrow$ Numbers
- Digital logic design allows for two states:
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 - ▶ On / Off (two voltage levels)
 - ▶ 1 / 0

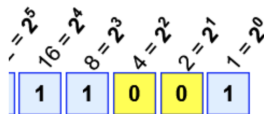
Binary Numbers

- Logic $\rightarrow$ Circuits $\rightarrow$ Numbers
- Digital logic design allows for two states:
 - ▶ True / False
 - ▶ On / Off (two voltage levels)
 - ▶ 1 / 0
- Computers store numbers using the Binary system (base 2)

Binary Numbers

- Logic → Circuits → Numbers
- Digital logic design allows for two states:
 - ▶ True / False
 - ▶ On / Off (two voltage levels)
 - ▶ 1 / 0
- Computers store numbers using the Binary system (base 2)
- A **bit** (binary digit) being 1 (on) or 0 (off)

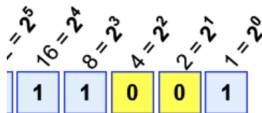
Binary Numbers



Example: $1 \times 16 + 1 \times 8 + 1 \times 1 = 16 + 8 + 1 = 25$

- Two digits: **0** and **1**

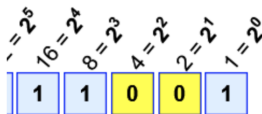
Binary Numbers



Example: $1 \times 16 + 1 \times 8 + 1 \times 1 = 16 + 8 + 1 = 25$

- Two digits: **0** and **1**
- Each position is a power of two

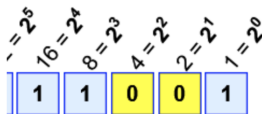
Binary Numbers



Example: $1 \times 16 + 1 \times 8 + 1 \times 1 = 16 + 8 + 1 = 25$

- Two digits: **0** and **1**
- Each position is a power of two
 - ▶ Decimal: the "ones", "tens", "hundreds" and so on (powers of 10)

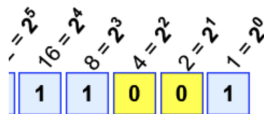
Binary Numbers



Example: $1 \times 16 + 1 \times 8 + 1 \times 1 = 16 + 8 + 1 = 25$

- Two digits: **0** and **1**
- Each position is a power of two
 - ▶ Decimal: the "ones", "tens", "hundreds" and so on (powers of 10)
 - ▶ Binary: the "ones", "twos", "fours", "sixteens" and so on (powers of 2)

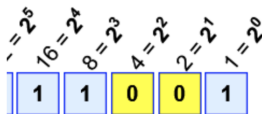
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- Two digits: **0** and **1**
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- In each position the digit is either 0 or 1, so given a binary number we can obtain the decimal equivalent as follows:

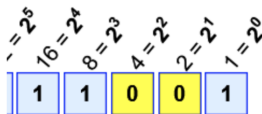
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 - ▶ Decimal: the "ones", "tens", "hundreds" and so on (powers of 10)
 - ▶ Binary: the "ones", "twos", "fours", "sixteens" and so on (powers of 2)
- In each position the digit is either 0 or 1, so given a binary number we can obtain the decimal equivalent as follows:
 - ▶ In the "ones" position we either have a 1 or not

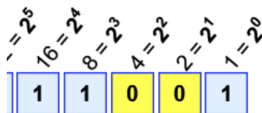
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- Two digits: **0** and **1**
- Each position is a power of two
 - ▶ Decimal: the "ones", "tens", "hundreds" and so on (powers of 10)
 - ▶ Binary: the "ones", "twos", "fours", "sixteens" and so on (powers of 2)
- In each position the digit is either 0 or 1, so given a binary number we can obtain the decimal equivalent as follows:
 - ▶ In the "ones" position we either have a 1 or not
 - ▶ In the "twos" position we either have a 2 or not

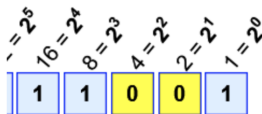
Binary Numbers



Example: $1 \times 16 + 1 \times 8 + 1 \times 1 = 16 + 8 + 1 = 25$

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 - ▶ Decimal: the "ones", "tens", "hundreds" and so on (powers of 10)
 - ▶ Binary: the "ones", "twos", "fours", "sixteens" and so on (powers of 2)
- In each position the digit is either 0 or 1, so given a binary number we can obtain the decimal equivalent as follows:
 - ▶ In the "ones" position we either have a 1 or not
 - ▶ In the "twos" position we either have a 2 or not
 - ▶ In the "fours" position we either have a 4 or not ...

Binary Numbers



Example: $1 \times 16 + 1 \times 8 + 1 \times 1 = 16 + 8 + 1 = 25$

- Two digits: **0** and **1**
- Each position is a power of two
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- In each position the digit is either 0 or 1, so given a binary number we can obtain the decimal equivalent as follows:
 - ▶ In the "ones" position we either have a 1 or not
 - ▶ In the "twos" position we either have a 2 or not
 - ▶ In the "fours" position we either have a 4 or not ...
- **Example:**

$$11001_{base2} = 16 + 8 + 1 = 25_{base10}$$

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- Write down the output to see the pattern:

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- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- Write down the output to see the pattern:

1

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- Write down the output to see the pattern:
1
2

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- Write down the output to see the pattern:

1

2

Fizz

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- Write down the output to see the pattern:

1

2

Fizz

4

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
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2

Fizz

4

Buzz

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Buzz

Fizz

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- Write down the output to see the pattern:

1

2

Fizz

4

Buzz

Fizz

7

Challenge: Tech Interview Classic

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- Write down the output to see the pattern:

1

2

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4

Buzz

Fizz

7

...

14

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
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1

2

Fizz

4

Buzz

Fizz

7

...

14

FizzBuzz

Challenge: Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- Write down the output to see the pattern:

1

2

Fizz

4

Buzz

Fizz

7

...

14

FizzBuzz

- Write the **algorithm** then, if time, write the code.

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.

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- To Do List:

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List:
 - ▶ Create a loop that goes from 1 to 100.

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List:
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ If the number is divisible by 3, print “Fizz”.

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List:
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ If the number is divisible by 3, print “Fizz”.
 - ▶ If the number is divisible by 5, print “Buzz”.

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 - ▶ Create a loop that goes from 1 to 100.
 - ▶ If the number is divisible by 3, print “Fizz”.
 - ▶ If the number is divisible by 5, print “Buzz”.
 - ▶ **If divisible by both, print “FizzBuzz”.**

Tech Interview Classic

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- To Do List:
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ If the number is divisible by 3, print “Fizz”.
 - ▶ If the number is divisible by 5, print “Buzz”.
 - ▶ **If divisible by both, print “FizzBuzz”.**
 - ▶ Otherwise print the number.

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List:
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ If the number is divisible by 3, print “Fizz”.
 - ▶ If the number is divisible by 5, print “Buzz”.
 - ▶ **If divisible by both, print “FizzBuzz”.**
 - ▶ Otherwise print the number.

Order matters!!! To print FizzBuzz when i is divisible by both it should be checked first, otherwise it will never get to this case!

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List (**Reordered**):

Tech Interview Classic

- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List (**Reordered**):
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ **If divisible by both 3 and 5, print “FizzBuzz”.**

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- Write a program that prints the numbers from 1 to 100. But for multiples of three print “Fizz” instead of the number and for the multiples of five print “Buzz”. For numbers which are multiples of both three and five print “FizzBuzz”.
- To Do List (**Reordered**):
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ **If divisible by both 3 and 5, print “FizzBuzz”.**
 - ▶ If the number is divisible by 3, print “Fizz”.
 - ▶ If the number is divisible by 5, print “Buzz”.
 - ▶ Otherwise print the number.
 - ▶ Also should print a new line (so each entry is on its own line).

Tech Interview Classic

- To Do List:
 - ▶ Create a loop that goes from 1 to 100.
 - ▶ If divisible by both 3 and 5, print “FizzBuzz”.
 - ▶ If the number is divisible by 3, print “Fizz”.
 - ▶ If the number is divisible by 5, print “Buzz”.
 - ▶ Otherwise print the number.
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Tech Interview Classic

- To Do List:

- ▶ Create a loop that goes from 1 to 100.
- ▶ If divisible by both 3 and 5, print “FizzBuzz”.
- ▶ If the number is divisible by 3, print “Fizz”.
- ▶ If the number is divisible by 5, print “Buzz”.
- ▶ Otherwise print the number.
- ▶ Also should print a new line (so each entry is on its own line).

```
for i in range(1,101):
```

Tech Interview Classic

- To Do List:

- ▶ Create a loop that goes from 1 to 100.
- ▶ If divisible by both 3 and 5, print “FizzBuzz”.
- ▶ If the number is divisible by 3, print “Fizz”.
- ▶ If the number is divisible by 5, print “Buzz”.
- ▶ Otherwise print the number.
- ▶ Also should print a new line (so each entry is on its own line).

```
for i in range(1,101):  
    if i%3 == 0 and i%5 == 0:  
        print("FizzBuzz")
```


Tech Interview Classic

- To Do List:

- ▶ Create a loop that goes from 1 to 100.
- ▶ If divisible by both 3 and 5, print “FizzBuzz”.
- ▶ If the number is divisible by 3, print “Fizz”.
- ▶ If the number is divisible by 5, print “Buzz”.
- ▶ Otherwise print the number.
- ▶ Also should print a new line (so each entry is on its own line).

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    elif i%3 == 0:  
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    elif i%5 == 0:  
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    else:  
        print(i)
```

Recap



- In Python, we introduced:

Recap



- In Python, we introduced:
 - ▶ Decisions
 - ▶ Logical Expressions
 - ▶ Circuits
 - ▶ Binary Numbers